



UNITED INTERNATIONAL UNIVERSITY

Department of Computer Science and Engineering (CSE)

Course Outline

1	Course Title	Computer Architecture						
2	Course Code	CSE 3313						
3	Trimester and Year	Spring 2026						
4	Pre-requisites	CSE 225 & CSE 226						
5	Credit Hours	3.00						
6	Section	A, E						
7	Class Hours	<table><tr><th>Section</th><th>Class Time</th></tr><tr><td>A</td><td>Sat, Tues: 9:51 AM to 11:10 AM</td></tr><tr><td>E</td><td>Sun, Wed: 9:51 to 11:10 AM</td></tr></table>	Section	Class Time	A	Sat, Tues: 9:51 AM to 11:10 AM	E	Sun, Wed: 9:51 to 11:10 AM
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8	Class Room	<table><tr><th>Section</th><th>Room Number</th></tr><tr><td>A</td><td>324</td></tr><tr><td>E</td><td>309</td></tr></table>	Section	Room Number	A	324	E	309
Section	Room Number							
A	324							
E	309							
9	Instructor's Name	Sidratul Tanzila Tasmi Lecturer, Dept. of CSE						
10	Email and Contact	tanzila@cse.uiu.ac.bd 01735944394						
11	Office	Room 319 (D)						
12	Counseling Hours	Sunday, Wednesday: 12:30 PM to 1:50 PM Tuesday: 11:11 AM to 1:50 PM						
13	Text Book	Computer Organization and Design: The Hardware/Software Interface, Fifth Edition, Patterson and Hennessy, Morgan Kaufmann/ Elsevier, 2014.						

14	Reference	<i>Introduction to Computing Systems: From Bits and Gates to C and Beyond</i> , Second Edition, Patt and Patel, McGraw-Hill. <i>Computer Organization</i> , Hamacher, Vranesic, and Zaky, McGraw-Hill. <i>Computer Architecture and Implementation</i> , Harvey Cragon, Cambridge University Press. <i>Structured Computer Organization</i> , Andrew Tanenbaum, Prentice Hall.						
15	Course Contents (approved by UGC)	Information representation and transfer, Instruction and data access methods, the control unit: hardwired and microprogrammed, memory organization, I/O systems, channels, interrupts, DMA, Von Neumann SISD organization, RISC and CISC machines. Pipelined machines, Interleaved memory system, Caches, Hardware and architectural issues of parallel machines, Array processors, associative processors, multiprocessors, systolic processors, data flow computers and interconnection networks, High level language concept of computer architecture.						
		Course Outcomes (COs)						
16		CO	Statement	Bloom's Domain	Program Outcome	Knowledge Profile	Complex Problem	Engineering

								g
								Activi
								ties
		CO1	Analyze the operations of a modern computer from software level to hardware level and explain instructions of	C	a	Engineering fundamentals (K3) & Engineering specialist knowledge(K4)	Depth of knowledge (P1)	-
			a MIPS based					
			instruction set					
			architecture					
		CO2	Understand the basic	C	a			
			of arithmetic operations of MIPS architecture		Engineering knowledge			
		CO3	Design pipelined and	C	b			
			non-pipelined processors for MIPS architecture		Problem analysis			
		CO4	Compare the usage of	C	b			

			different types of interleaved memory		Problem analysis			
17	Teaching Methods	Lecture, Case Studies, Project Developments.						

18	Assessment Method																																																
	<table><tr><td colspan="2">Assessment Criteria</td><td colspan="5">Percentage (%)</td></tr><tr><td colspan="2">Attendance</td><td colspan="5">5</td></tr><tr><td colspan="2">Class Tests</td><td colspan="5">20</td></tr><tr><td colspan="2">Assignment</td><td colspan="5">5</td></tr><tr><td colspan="2">Mid Term</td><td colspan="5">30</td></tr><tr><td colspan="2">Final</td><td colspan="5">40</td></tr></table>							Assessment Criteria		Percentage (%)					Attendance		5					Class Tests		20					Assignment		5					Mid Term		30					Final		40				
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19	Lecture Outline																																																
		Class	Topics/Assignments	COs	Reading Reference	Lecture Outcomes/Activities																																											
		1	Introduction to Computer Architecture	CO1	1.1																																												

2	Components of Computer, Performance of Processor	CO1	1.2, 1.3, 1.4	
3	Power Wall, Multiprocessor, Pitfalls, Amdahl's Law	CO1	1.5, 1.6, 1.8	
4	Introduction to ISA, Operations of Computer, Operands of Computer	CO1	2.1, 2.2, 2.3	Class Test 1
5	Signed & Unsigned number, Representing Instructions	CO1	2.4, 2.5	
6	Logical Operations, Branching Instructions	CO1	2.6, 2.7	Assignment 1
7	Procedures	CO1	2.8	

8	Translating a Program, Fallacies and pitfalls	C01	2.12, 2.18	
9	Arithmetic of Computer, Addition & Subtraction	C02	3.1, 3.2	Class Test 2
10	Multiplication Algorithm, Optimized Multiplication Algorithm	C02	3.3	
11	Division Algorithm & Modified Division Algorithm	C02	3.4	
12	Review of Mid Term Syllabus	-		
	MIDTERM EXAM			
13	Floating Point, Floating Point Addition	C02	3.5	Assignment 2
14	Logic Design Convention, Building a Datapath	C03	4.1, 4.2, 4.3	
15	Simple Non-Pipelined Implementation of MIPS Datapath	C03	4.4	
16	Pipelining Overview	C03	4.5	Class Test 3
17	Pipeline Hazards	C03	4.5	
18	Pipelined Datapath & Control	C03	4.6	
19	Forwarding vs Stalling, Branch Prediction	C03	4.7, 4.8	Class Test 4
20	Introduction to Memory Hierarchy, Basics of Cache	C04	5.1, 5.2	
21	Cache Miss, Cache performance	C04	5.2, 5.3	
22	Virtual Memory	C04	5.4	Class Test 5
23	Page Faults, Address Translation	C04	5.4	
24	Review of Final Syllabus	-		

Appendix 1: Assessment Metho

Assessment Types	Marks
Attendance	5%
Assignments	5%
Class Tests	20%
Mid Term	30%
Final Exam	40%

Appendix 2: Grading Policy

Letter Grade	Marks %	Grade Point	Letter Grade	Marks%	Grade Point
A (Plain)	90-100	4.00	C+ (Plus)	70-73	2.33
A- (Minus)	86-89	3.67	C (Plain)	66-69	2.00
B+ (Plus)	82-85	3.33	C- (Minus)	62-65	1.67
B (Plain)	78-81	3.00	D+ (Plus)	58-61	1.33
B- (Minus)	74-77	2.67	D (Plain)	55-57	1.00
			F (Fail)	<55	0.00

Appendix-3: Program outcomes

POs	Program Outcomes
PO a	Engineering knowledge
PO b	Problem analysis
PO c	Design/ development of solutions
PO d	Investigation: Engagement with selected knowledge in the research literature of the discipline
PO e	Modern Tool Usage: Level of understanding of the appropriateness of the tool
PO f	The Engineer and Society engineering practice, ethics and the professional responsibility, public safety; economic, social, cultural, environmental and sustainability
PO g	Environment and Sustainability: Type of solutions.
PO h	Ethics: Understanding and level of practice
PO i	Individual and Team work: Role in and diversity of team
PO j	Communication: Level of communication according to type of activities performed
PO k	Project Management and Finance: Level of management required for differing types of activity
PO l	Lifelong learning: Preparation for and depth of continuing learning.